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TECNOLÓGICO
NACIONAL DE MÉXICO



2026. "Año del Humanismo Mexicano en el Estado de México"

Tecnológico de estudios Superiores de Villa Guerrero.

Ingeniería en Sistemas Computacionales

UNIDAD DE APRENDIZAJE:

SISTEMAS OPERATIVOS

PRÁCTICA:

PRÁCTICA 3. "EXPLORANDO EL HARDWARE DE TU SISTEMA CON LSPCI"

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OCTUBRE DE 2026



Introducción

En esta práctica se explorará el hardware de un sistema Linux mediante el uso del comando `lspci`, una herramienta que permite identificar los dispositivos conectados al equipo a través del bus PCI (Peripheral Component Interconnect). Este comando proporciona información sobre los diferentes componentes y periféricos que forman parte del sistema, permitiendo conocer de manera más detallada el hardware que está siendo reconocido por el sistema operativo.

A lo largo de la práctica se ejecutará inicialmente `lspci` sin opciones para obtener una lista básica de los dispositivos detectados. Posteriormente, se utilizará la opción `-v`, la cual permite obtener información más detallada de estos dispositivos, incluyendo datos relacionados con los controladores o módulos del kernel que están siendo utilizados. De esta manera, se podrá analizar el hardware del equipo y reconocer qué controladores utiliza tanto la tarjeta de red inalámbrica como la tarjeta gráfica.

Objetivo

Familiarizarse con el comando `lspci` para identificar y entender los dispositivos de hardware (periféricos) conectados a tu sistema a través del bus PCI (Peripheral Component Interconnect).

Requisitos

1. Sistema Linux (Debian, Ubuntu, Arch, etc.)
2. Acceso a terminal



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1. lspci

Muestra una lista de los dispositivos de hardware que están conectados y son reconocidos por el sistema mediante el bus PCI.

```
user@debian:~$ lspci
0000:00:00.0 Host bridge: Intel Corporation Device a719 (rev 01)
0000:00:01.0 PCI bridge: Intel Corporation Raptor Lake PCI Express 5.0 Graphics Port (PEG010) (rev 01)
0000:00:02.0 VGA compatible controller: Intel Corporation Raptor Lake-S UHD Graphics (rev 04)
0000:00:04.0 Signal processing controller: Intel Corporation Raptor Lake Dynamic Platform and Thermal Framework Processor Participant (rev 01)
0000:00:08.0 System peripheral: Intel Corporation GNA Scoring Accelerator module (rev 01)
0000:00:0e.0 RAID bus controller: Intel Corporation Volume Management Device NVMe RAID Controller Intel Corporation
0000:00:14.0 USB controller: Intel Corporation Raptor Lake USB 3.2 Gen 2x2 (20 Gb/s) XHCI Host Controller (rev 11)
0000:00:14.2 RAM memory: Intel Corporation Raptor Lake-S PCI Shared SRAM (rev 11)
0000:00:15.0 Serial bus controller: Intel Corporation Raptor Lake Serial IO I2C Host Controller #0 (rev 11)
0000:00:15.1 Serial bus controller: Intel Corporation Raptor Lake Serial IO I2C Host Controller #1 (rev 11)
0000:00:16.0 Communication controller: Intel Corporation Raptor Lake CME HECI #1 (rev 11)
0000:00:19.0 Serial bus controller: Intel Corporation Raptor Lake Serial IO I2C Host Controller (rev 11)
0000:00:1a.0 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port #25 (rev 11)
0000:00:1c.0 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port (rev 11)
0000:00:1c.5 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port (rev 11)
0000:00:1f.0 ISA bridge: Intel Corporation HM770 Chipset LPC/aSPI Controller (rev 11)
0000:00:1f.3 Audio device: Intel Corporation Raptor Lake High Definition Audio Controller (rev 11)
0000:00:1f.4 SMBus: Intel Corporation Raptor Lake-S PCI SMBus Controller (rev 11)
0000:00:1f.5 Serial bus controller: Intel Corporation Raptor Lake SPI (Flash) Controller (rev 11)
0000:02:00.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge (Maple Ridge 2C 2020) (rev 02)
0000:03:00.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge (Maple Ridge 2C 2020) (rev 02)
0000:03:01.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge (Maple Ridge 2C 2020) (rev 02)
0000:03:02.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge (Maple Ridge 2C 2020) (rev 02)
0000:03:03.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge (Maple Ridge 2C 2020) (rev 02)
0000:04:00.0 USB controller: Intel Corporation Thunderbolt 4 NHI (Maple Ridge 2C 2020)
0000:38:00.0 USB controller: Intel Corporation Thunderbolt 4 USB Controller (Maple Ridge 2C 2020)
0000:6c:00.0 Network controller: Realtek Semiconductor Co., Ltd. RTL8852CE PCIe 802.11ax Wireless Network Controller (rev 01)
0000:6d:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168/8211/8411 PCI Express Gigabit Ethernet Controller (rev 15)
10000:e0:1d.0 System peripheral: Intel Corporation RST VMD Managed Controller
10000:e0:1d.4 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port (rev 11)
10000:e1:00.0 Non-Volatile memory controller: Sandisk Corp WD PC SN5000S M.2 2280 NVMe SSD (DRAM-less)
user@debian:~$
```

2. lspci -v

Muestra los dispositivos PCI con información más detallada, incluyendo datos del dispositivo y el controlador (driver) que está utilizando.

```
user@debian:~$ lspci -v
0000:00:00.0 Host bridge: Intel Corporation Device a719 (rev 01)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: bus master, fast devsel, latency 0, IOMMU group 1

0000:00:01.0 PCI bridge: Intel Corporation Raptor Lake PCI Express 5.0 Graphics Port (PEG010) (rev 01) (prog-if 00 [Normal decode])
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 3069
Flags: bus master, fast devsel, latency 0, IRQ 122, IOMMU group 2
Bus: primary00, secondary=01, subordinate=01, sec-latency=0
I/O behind bridge: 5000-5fff [size=4K] [16-bit]
Memory behind bridge: 8a000000-8b0fffff [size=0M] [32-bit]
Prefetchable memory behind bridge: 7400000000-7601ffffff [size=0224M] [32-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:00:02.0 VGA compatible controller: Intel Corporation Raptor Lake-S UHD Graphics (rev 04) (prog-if 00 [VGA controller])
DeviceName: Onboard - Video
Subsystem: ASUSTek Computer Inc. Device 3069
Flags: bus master, fast devsel, latency 0, IRQ 202, IOMMU group 0
Memory at 7604000000 (64-bit, non-prefetchable) [size=16M]
Memory at 4000000000 (64-bit, prefetchable) [size=250M]
I/O ports at 6000 [size=64]
Expansion ROM at 900c0000 [virtual] [disabled] [size=128K]
Capabilities: <access denied>
Kernel driver in use: i915
Kernel modules: i915, xe

0000:00:04.0 Signal processing controller: Intel Corporation Raptor Lake Dynamic Platform and Thermal Framework Processor Participant (rev 01)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: bus master, fast devsel, latency 0, IRQ 10, IOMMU group 3
Memory at 7605200000 (64-bit, non-prefetchable) [size=128K]
Capabilities: <access denied>
Kernel driver in use: proc_thermal_pci
Kernel modules: processor_thermal_device_pci

0000:00:08.0 System peripheral: Intel Corporation GNA Scoring Accelerator module (rev 01)
DeviceName: Onboard - Other
Flags: fast devsel, IRQ 255, IOMMU group 4
Memory at 760523e000 (64-bit, non-prefetchable) [disabled] [size=4K]
Capabilities: <access denied>

0000:00:0e.0 RAID bus controller: Intel Corporation Volume Management Device NVMe RAID Controller Intel Corporation (prog-if 00 [Vendor specific])
DeviceName: Onboard - Other
Subsystem: Intel Corporation Device 0000
Flags: bus master, fast devsel, latency 0, IOMMU group 5
Memory at 7602000000 (64-bit, non-prefetchable) [size=32M]
Memory at 8a000000 (32-bit, non-prefetchable) [size=32M]
Memory at 7605100000 (64-bit, non-prefetchable) [size=1M]
Capabilities: <access denied>
Kernel driver in use: vmd
Kernel modules: vmd

0000:00:14.0 USB controller: Intel Corporation Raptor Lake USB 3.2 Gen 2x2 (20 Gb/s) XHCI Host Controller (rev 11) (prog-if 30 [XHCI])
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 201f
Flags: bus master, medium devsel, latency 0, IRQ 167, IOMMU group 6
Memory at 7605200000 (64-bit, non-prefetchable) [size=64K]
Capabilities: <access denied>
Kernel driver in use: xhci_hcd
Kernel modules: mei_me, xhci_pci
```



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```
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user@debian:~$ lsusb -v

0000:00:14.2 RAM memory: Intel Corporation Raptor Lake-S PCH Shared SRAM (rev 11)
DeviceName: Onboard - Other
Flags: fast devel, IOMMU group 8
Memory at 7605234000 (64-bit, non-prefetchable) [disabled] [size=16K]
Memory at 7605234000 (64-bit, non-prefetchable) [disabled] [size=4K]
Capabilities: <access denied>

0000:00:15.0 Serial bus controller: Intel Corporation Raptor Lake Serial IO I2C Host Controller #0 (rev 11)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: bus master, fast devel, latency 0, IRQ 27, IOMMU group 7
Memory at 4017000000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: intel-lpss
Kernel modules: intel_lpss_pci

0000:00:15.1 Serial bus controller: Intel Corporation Raptor Lake Serial IO I2C Host Controller #1 (rev 11)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: bus master, fast devel, latency 0, IRQ 40, IOMMU group 7
Memory at 4017000000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: intel-lpss
Kernel modules: intel_lpss_pci

0000:00:16.0 Communication controller: Intel Corporation Raptor Lake CSME HECI #1 (rev 11)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: bus master, fast devel, latency 0, IRQ 203, IOMMU group 8
Memory at 7605234000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: mei_me
Kernel modules: mei_me

0000:00:19.0 Serial bus controller: Intel Corporation Raptor Lake Serial IO I2C Host Controller (rev 11)
DeviceName: Onboard - Other
Flags: bus master, fast devel, latency 0, IRQ 31, IOMMU group 9
Memory at 4017002000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: intel-lpss
Kernel modules: intel_lpss_pci

0000:00:1a.0 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port #25 (rev 11) (prog-if 00 [Normal decode])
Flags: bus master, fast devel, latency 0, IRQ 123, IOMMU group 10
Bus: primary=00, secondary=02, subordinate=00, sec-latency=0
I/O behind bridge: 7000-9fff [size=12K] [16-bit]
Memory behind bridge: 54000000-820fffff [size=737M] [32-bit]
Prefetchable memory behind bridge: 6000000000-72000fffff [size=73729M] [32-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:00:1c.0 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port (rev 11) (prog-if 00 [Normal decode])
Flags: bus master, fast devel, latency 0, IRQ 124, IOMMU group 11
Bus: primary=00, secondary=0c, subordinate=0c, sec-latency=0
I/O behind bridge: 4000-4fff [size=4K] [16-bit]
Memory behind bridge: 8c100000-8c1fffff [size=1M] [32-bit]
Prefetchable memory behind bridge: [disabled] [64-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:00:1c.5 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port (rev 11) (prog-if 00 [Normal decode])
Flags: bus master, fast devel, latency 0, IRQ 125, IOMMU group 12
Bus: primary=00, secondary=0d, subordinate=0d, sec-latency=0
I/O behind bridge: 3000-3fff [size=4K] [16-bit]
Memory behind bridge: 8c000000-8c0fffff [size=1M] [32-bit]
Prefetchable memory behind bridge: [disabled] [64-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:00:1f.0 ISA bridge: Intel Corporation HM770 Chipset LPC/eSPI Controller (rev 11)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: bus master, fast devel, latency 0, IOMMU group 13

0000:00:1f.3 Audio device: Intel Corporation Raptor Lake High Definition Audio Controller (rev 11) (prog-if 00 [HDA compatible with vendor specific extensions])
DeviceName: Onboard - Sound
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: bus master, fast devel, latency 32, IRQ 204, IOMMU group 13
Memory at 7605234000 (64-bit, non-prefetchable) [size=16K]
Memory at 7605060000 (64-bit, non-prefetchable) [size=1M]
Capabilities: <access denied>
Kernel driver in use: snd_hda_intel
Kernel modules: snd_hda_intel, snd_soc_avs, snd_soc_avs, snd_soc_pci_intel_tpl

0000:00:1f.4 SMBus: Intel Corporation Raptor Lake-S PCH SMBus Controller (rev 11)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: medium devel, IRQ 18, IOMMU group 13
Memory at 7605234000 (64-bit, non-prefetchable) [size=256]
I/O ports at efa0 [size=32]
Kernel driver in use: i801_smbus
Kernel modules: i2c_i801

0000:00:1f.5 Serial bus controller: Intel Corporation Raptor Lake SPI (flash) Controller (rev 11)
DeviceName: Onboard - Other
Subsystem: ASUSTek Computer Inc. Device 1004
Flags: fast devel, IOMMU group 13
Memory at 52000000 (32-bit, non-prefetchable) [size=4K]

0000:02:00.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge [Maple Ridge 2C 2020] (rev 02) (prog-if 00 [Normal decode])
Subsystem: SpacemIT Device 1043
Physical Slot: 28
Flags: bus master, fast devel, latency 0, IRQ 16, IOMMU group 14
Bus: primary=02, secondary=03, subordinate=0b, sec-latency=0
I/O behind bridge: 7000-8fff [size=8K] [16-bit]
Memory behind bridge: 54000000-81efffff [size=739M] [32-bit]
Prefetchable memory behind bridge: 6000000000-72000fffff [size=73729M] [32-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:03:00.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge [Maple Ridge 2C 2020] (rev 02) (prog-if 00 [Normal decode])
Subsystem: SpacemIT Device 1043
Flags: bus master, fast devel, latency 0, IRQ 126, IOMMU group 15
Bus: primary=03, secondary=04, subordinate=04, sec-latency=0
I/O behind bridge: [disabled] [32-bit]
Memory behind bridge: [disabled] [32-bit]
Prefetchable memory behind bridge: 7200000000-72000fffff [size=1M] [32-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:03:01.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge [Maple Ridge 2C 2020] (rev 02) (prog-if 00 [Normal decode])
.....
```



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```
Oct 4 3:48 PM
user@debian:~$ lsusb
0000:03:01.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge [Maple Ridge 2C 2020] (rev 02) (prog-if 00 [Normal decode])
Subsystem: SpacemIT Device 1043
Flags: bus master, fast devsel, latency 0, IRQ 127, IOMMU group 16
Bus: primary=03, secondary=09, subordinate=37, sec-latency=0
I/O behind bridge: 7000-7fff [size=4K] [16-bit]
Memory behind bridge: 6af00000-81dfffff [size=367M] [32-bit]
Prefetchable memory behind bridge: 600000000-71ffffff [size=36G] [32-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:03:02.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge [Maple Ridge 2C 2020] (rev 02) (prog-if 00 [Normal decode])
Subsystem: SpacemIT Device 1043
Flags: bus master, fast devsel, latency 0, IRQ 128, IOMMU group 17
Bus: primary=03, secondary=30, subordinate=30, sec-latency=0
I/O behind bridge: [disabled] [32-bit]
Memory behind bridge: 81e00000-81e1ffff [size=1M] [32-bit]
Prefetchable memory behind bridge: [disabled] [64-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:03:03.0 PCI bridge: Intel Corporation Thunderbolt 4 Bridge [Maple Ridge 2C 2020] (rev 02) (prog-if 00 [Normal decode])
Subsystem: SpacemIT Device 1043
Flags: bus master, fast devsel, latency 0, IRQ 129, IOMMU group 18
Bus: primary=03, secondary=39, subordinate=60, sec-latency=0
I/O behind bridge: 8000-8fff [size=4K] [16-bit]
Memory behind bridge: 54000000-6aefffff [size=367M] [32-bit]
Prefetchable memory behind bridge: 600000000-68ffffff [size=36G] [32-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

0000:04:00.0 USB controller: Intel Corporation Thunderbolt 4 NHI [Maple Ridge 2C 2020] (prog-if 40 [USB4 Host Interface])
Subsystem: SpacemIT Device 1043
Flags: bus master, fast devsel, latency 0, IRQ 16, IOMMU group 19
Memory at 7200000000 (64-bit, prefetchable) [size=256K]
Memory at 7200040000 (64-bit, prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: thunderbolt
Kernel modules: thunderbolt

0000:30:00.0 USB controller: Intel Corporation Thunderbolt 4 USB Controller [Maple Ridge 2C 2020] (prog-if 30 [XHCI])
Flags: bus master, fast devsel, latency 0, IRQ 175, IOMMU group 20
Memory at 81e00000 (64-bit, non-prefetchable) [size=64K]
Capabilities: <access denied>
Kernel driver in use: xhci_hcd
Kernel modules: xhci_pci

0000:6c:00.0 Network controller: Realtek Semiconductor Co., Ltd. RTL8852CE PCIe 802.11ax Wireless Network Controller (rev 01)
Subsystem: AzureWave Device 5603
Flags: bus master, fast devsel, latency 0, IRQ 205, IOMMU group 21
I/O ports at 4000 [size=256]
Memory at 8c000000 (64-bit, non-prefetchable) [size=1M]
Memory at 8c000000 (64-bit, non-prefetchable) [size=16K]
Capabilities: <access denied>
Kernel driver in use: rtw89_8852ce
Kernel modules: rtw89_8852ce

0000:6d:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168/8211/8411 PCI Express Gigabit Ethernet Controller (rev 15)
Subsystem: ASUSTek Computer Inc. Device 208f
Flags: bus master, fast devsel, latency 0, IRQ 17, IOMMU group 22
I/O ports at 3000 [size=256]
Memory at 8c004000 (64-bit, non-prefetchable) [size=4K]
Memory at 8c000000 (64-bit, non-prefetchable) [size=16K]
Capabilities: <access denied>
Kernel driver in use: r8169
Kernel modules: r8169

0000:6d:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168/8211/8411 PCI Express Gigabit Ethernet Controller (rev 15)
Subsystem: ASUSTek Computer Inc. Device 208f
Flags: bus master, fast devsel, latency 0, IRQ 17, IOMMU group 22
I/O ports at 3000 [size=256]
Memory at 8c004000 (64-bit, non-prefetchable) [size=4K]
Memory at 8c000000 (64-bit, non-prefetchable) [size=16K]
Capabilities: <access denied>
Kernel driver in use: r8169
Kernel modules: r8169

10000:e0:1d.0 System peripheral: Intel Corporation RST VMD Managed Controller
Flags: fast devsel, IOMMU group 5

10000:e0:1d.4 PCI bridge: Intel Corporation Raptor Lake PCI Express Root Port (rev 11) (prog-if 00 [Normal decode])
Flags: bus master, fast devsel, latency 0, IRQ 104, IOMMU group 5
Bus: primary=00, secondary=e1, subordinate=e1, sec-latency=0
I/O behind bridge: [disabled] [16-bit]
Memory behind bridge: 8a000000-8a01ffff [size=1M] [32-bit]
Prefetchable memory behind bridge: [disabled] [64-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

10000:e1:00.0 Non-Volatile memory controller: Sandisk Corp WD PC SM5000S M.2 2280 NVMe SSD (DRAM-less) (prog-if 02 [NVM Express])
Subsystem: Sandisk Corp WD PC SM5000S M.2 2280 NVMe SSD (DRAM-less)
Flags: bus master, fast devsel, latency 0, IOMMU group 5
BIOS result: 00
Memory at 8a000000 (64-bit, non-prefetchable) [size=16K]
Capabilities: <access denied>
Kernel driver in use: nvme
Kernel modules: nvme

user@debian:~$
```

Preguntas

- ¿Qué driver está usando tu tarjeta de red Wi-Fi (Wireless)? rtw89_8852ce
- ¿Qué driver está usando tu tarjeta gráfica? i915



Conclusión

Al finalizar la práctica se logró conocer el funcionamiento del comando `lspci` y su utilidad para identificar los dispositivos de hardware reconocidos por un sistema Linux. Mediante el comando básico fue posible obtener una lista de los componentes conectados al bus PCI, mientras que con la opción `-v` se pudo consultar información más detallada sobre los dispositivos y los controladores utilizados por el sistema.

Esta práctica permitió comprender que la terminal de Linux proporciona herramientas que facilitan la identificación y consulta del hardware de un equipo sin necesidad de utilizar aplicaciones gráficas. Además, conocer los controladores asociados a dispositivos como la tarjeta Wi-Fi y la tarjeta gráfica resulta útil para comprender cómo el sistema operativo reconoce y utiliza los diferentes componentes de hardware.

